CDFD Runtime Paper 12: Candidate Laws, Discovery Program, and Responsible Use

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Abstract

Executive synthesis. Earlier drafts called the final synthesis "immutable laws." This upgraded paper uses more disciplined language: candidate laws, model invariants, and falsification targets. CDFD Runtime experiments suggest recurring motifs–adaptive constraint, operating-ratio limits, structural memory, weakest-link throughput, and Life Number thresholds. These motifs are useful research guides, not final proof that reality has been fully decoded.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α, β, and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η, λ, μ, ν, ρ, σ, τ, and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Why the Name Changes

The phrase "immutable laws" overstates the current evidence. The runtime has scripts, diagnostics, and reports. It does not yet have enough independent empirical validation to call every recurrent pattern a universal law. The final paper therefore names a research program: candidate Mujjabi laws that remain open to testing. Named CLI-facing tests are listed in `MUJJABI_RUNTIME_LAWS_AND_TESTS.md`, including aromatic source-mix and photochemical endpoint guardrails added in the May 2026 runtime pass.

3 Candidate Mujjabi Law 1: Adaptive Constraint

Sustained flow tends to generate, reveal, or amplify constraint. In runtime form, this appears through the constraint equation

$$\partial_t C = \alpha|\partial_t \Phi| - \beta C + \gamma \nabla^2 C. \tag{1}$$

The falsification target is simple: find mapped systems where increasing drive does not produce any measurable constraint response and where that absence is not already explained by a saturated or buffered medium.

4 Candidate Mujjabi Law 2: Operating-Ratio Regimes

The relation between Φ and C predicts regime transitions. The operating ratio

$$\Psi_s = \frac{\Phi}{C} S M_s \tag{2}$$

is the central runtime diagnostic. It must be validated separately in each domain. A high Ψ_s in one domain may mean overload; in another it may mean transient productive growth. Interpretation is not automatic.

5 Candidate Mujjabi Law 3: Structural Memory

Overload can leave retained memory in the medium. The vacuum-hysteresis experiment selected in `FINAL_RELEASE_EXPERIMENT_SELECTION.md` records a localized M_s residual. This supports the runtime's memory mechanism as a candidate diagnostic and a measurement protocol. It does not prove a physical vacuum-memory field.

6 Candidate Mujjabi Law 4: Weakest-Link Throughput

Network throughput is often limited by its most constrained effective channel. This appears in origins-of-life, medicine, networks, infrastructure, and supply chains. The generic diagnostic is useful, but real systems require real capacity data. The law fails where redundancy, storage, or rerouting invalidates the simple bottleneck approximation.

7 Candidate Mujjabi Law 5: Life Number Threshold

Sustained life-like throughput requires energy input, electron transport, proton coupling, relaxation, and stabilization to balance. The Life Number is

$$\Lambda = \frac{\Phi \sigma_e \sigma_p \tau_{\mathrm{relax}}}{S_{\mathrm{stab}} E_{\mathrm{maintenance}}}. \tag{3}$$

The OOL phase experiment treats $\Lambda = 1$ as a candidate transition boundary. The claim remains model-bound until laboratory and geochemical tests challenge the mapping.

8 Discovery Program

The local discovery program now has four evidence levels:

1. selected outputs such as the Mujjabi Vacuum Hysteresis Test, the Mujjabi $n = 7$ Knot Stability Test, and the Mujjabi Life Number Boundary Test;
2. broad sweeps such as frontier and gigamarathon outputs;
3. reports that define candidate interpretations and falsification conditions;
4. CLI and CDFL commands that make future runs reproducible.

9 Responsible Use

No runtime output is medical advice, device certification, financial policy, or physical proof without external validation. CDFD is a framework for asking sharper questions. Its power increases when its claims are clearly bounded.

Optional LLM provider interpretation does not change that boundary. It may summarize or question a saved deterministic result, but it is not part of the engine, not a validator, and not a substitute for domain evidence.

10 Conclusion

The synthesis of the runtime series is not certainty. It is a disciplined research engine: define variables, run the model, save the output, expose the failure condition, and let better evidence decide which candidate laws survive.

References

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[6] Steve Bico Mujjabi, MD. *CDFD Part II: Origins of Life and Tri-Regime Bioenergetics.* Public release archive, 2026, `https://doi.org/10.5281/zenodo.20264779`.